Personalizing Automated Continuing Care Support Messaging for Alcohol Use Disorder

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Abstract

Alcohol use disorder (AUD) and other substance use disorders (SUDs) are chronic conditions with high relapse rates and limited access to continuing care. Automated participant monitoring and support systems (APMSSs) offer a promising solution by using smartphones to monitor symptoms, predict relapse risk, and deliver personalized interventions. Curtin and colleagues have demonstrated the feasibility of smartphone-based ecological momentary assessments (EMAs) and machine learning models to predict daily relapse risk. However, effective implementation requires optimizing how participants receive and engage with these automated messages. This study evaluates various supportive language tones: legitimization of distress, acknowledgment of feelings, self-efficacy promotion, value affirmation, and promotion of social norms, and their impact on participants' perceptions of helpfulness in daily messages from an APMSS. 510 participants in early recovery from AUD completed a Qualtrics panel study containing personalized messages generated by a large language model (LLM) based on simulated user contexts and language characteristics. Participants rated these messages on perceived helpfulness, generally finding them to be helpful. Findings will inform best practices for delivering artificial intelligence (AI)-driven supportive messaging and optimizing automated continuing care for individuals in recovery from SUDs.

Keywords: automated systems, substance use disorder, computers as social actors, messaging styles, healthcare messaging, supportive language, perception of helpfulness in automated messaging

Introduction

Alcohol and other substance use disorders (SUDs) are serious chronic conditions, characterized by high relapse rates (McLellan et al., 2000; Dennis & Scott, 2007), substantial co-morbidity with other physical and mental health problems (Dennis & Scott, 2007; Substance Abuse and Mental Health Services Administration, n.d.), and an increased risk of mortality (Hedegaard et al., n.d.; Centers for Disease Control and Prevention (CDC), n.d.). Too few individuals receive medications or clinician-delivered interventions to help them initially achieve abstinence and/or reduce harms associated with their use (Substance Abuse and Mental Health Services Administration, n.d.), and even fewer receive continuing care. Continuing care, including both risk monitoring and ongoing support, is the gold standard for managing chronic health conditions such as diabetes, asthma, and HIV. Yet continuing care for SUDs is largely lacking despite ample evidence that SUDs are chronic, relapsing conditions (Substance Abuse and Mental Health Services Administration, n.d.; Stanojlović & Davidson, 2021; Socías et al. 2016).

Clinical researchers are now in the early stages of developing automated participant monitoring and support systems (APMSSs) to address this unmet need for SUD continuing care (McKay, 2021; Wyant et al., 2024). These APMSSs can deliver scalable and cost-efficient personalized care by using smartphones to both monitor participants' symptoms and relapse risks and deploy algorithms that recommend interventions and other supports that precisely target those symptoms and risks each day. Curtin and colleagues have demonstrated that smartphone-based ecological momentary assessments (EMAs; i.e., brief surveys completed each day on individuals' phones) are feasible and effective for long-term (the study examined a maximum duration of a year) monitoring of individuals with an opioid use disorder (Wyant et al., 2023; Moshontz et al., 2021). Furthermore, they have developed an algorithm that can accurately predict future lapses back to substance use with high temporal specificity (i.e., lapses within the next day) using these EMAs as inputs (Wyant et al., 2024). These same algorithms can also identify the most important risk factors for a lapse during that next day, which can be used to recommend daily interventions or other supports specific to that participant.

The APMSSs developed by Curtin and colleagues and others (Wyant et al., 2024; Moshontz et al., 2021; Gonzales et al., 2016; Haug et al., 2015) have enormous potential to fill the gap in continuing care for SUD. These systems can provide continuous risk monitoring and deliver daily personalized messaging to participants to guide them in addressing barriers to their recovery. The algorithms created by Curtin et al. can predict lapse risk probability, identify key contributing risk factors, and recommend interventions daily for each individual. However, to successfully implement these APMSSs for clinical benefits, we must first determine how best to provide these algorithm outputs in a helpful and personally relevant way to individuals in recovery.

There is substantial evidence that the language used in supportive and other advice messaging is crucial for participants to trust and follow advice when it is provided by healthcare professionals (e.g., doctors) in healthcare and similar settings. Classic research by Burleson and others has suggested that effective supportive advice messaging should 1) validate felt distress, 2) express caring, and 3) acknowledge participants' feelings (Burleson & Samter, 1985; MacGeorge et al., 2017). Advice messages that validate felt distress convey that the participant’s feelings are legitimate and do not criticize or punish these feelings, but rather acknowledge them as a step towards fixing the problem (Burleson & Samter, 1985; Burleson, 2003). Messages that express caring convey positive intentions, willingness to cooperate with the participant’s goals, acceptance of the participant, and availability to help (Burleson, 2003; Cramer, 1990; Sullivan, 1996). Messages that acknowledge feelings convey interest in listening to the participant, encourage them to express and explore their emotions, hypothesize about their feelings (e.g. "I bet that was difficult" or "I imagine you felt …"), and restate or reflect feelings (Burleson & Samter, 1985; Burleson, 2003; Barbee, 1990; Barbee & Cunningham, 1995; Burleson, 1994).

There are additional message components that are supported by more contemporary literature. Messages that promote self-efficacy have been found to improve participants' confidence and boost intrinsic motivation towards recovery (Lu et al., 2025). Messages that encourage self-affirmation of personal values have been found to reduce defensiveness towards receiving critical health information and promote acceptance, intent to change and behavior improvement (Epton et al., 2015). Social norms messaging can assist participants in recovery by providing "social proof" that recovery methods are effective (Papakonstantinou et al., 2025). Advice messaging that includes language consistent with each of these dimensions has been demonstrated to increase participant receptiveness and engagement with the advice provided to them (MacGeorge et al., 2017).

In addition to the language tones used in these messages, formality style matters. Messages written in a professional, formal style are judged to be more credible, authoritative and competent. Research supports the notion that formal messages framed as coming from a professional increase perceived competence of the source of assistance and helps participants accept the messages they are receiving and encourage them to take action based on the content of these messages (Linos et al., 2024). By contrast, messages written in an informal style, as would be expected to be received from a friend or peer, promote friendliness and credibility when delivered by a generative artificial intelligence (AI) chatbot (Zhao et al., 2024).

This rich literature on supportive advice messaging provides us with a strong foundation on which to craft daily messaging to participants from an APMSS. However, it is not clear how these tones and styles of messages will generalize to a context where messages are received from an automated system rather than a human advisor (e.g., a doctor). Participants may view any message delivered by an APMSS as less helpful than when it is delivered by a human healthcare professional, particularly when messages rely on language characteristics that seem inherently human (e.g., empathy, caring). Conversely, the Computers as Social Actors paradigm (Nass, Steuer & Tauber, 1994) suggests when algorithm-based technologies present humanlike attributes, users will perceive these technologies as social actors and transfer human-human communication scripts to human-technology interaction, suggesting that humanlike characteristics may be beneficial for some individuals (Edwards et al., 2019; Carolus et al., 2018).

Given this uncertainty, the proposed research will directly manipulate and evaluate the impact of each of five message tones (legitimizing, self-efficacy, acknowledging, value affirmation and norms) and each of two message styles (formal and informal) on participants' perceptions of the helpfulness offered by daily messages from an APMSS. The "caring" construct overlaps conceptually with other tones and has interpretive challenges in automated contexts. We have three goals for this study: 1) develop a system to use LLMs to write helpful daily support messages in various tones and styles, 2) develop a machine learning model to predict message helpfulness from demographics, preferences, tone, and style, and 3) compare the use of the machine learning model to other strategies to select the best tones and styles for future use. The results of this research will inform creators of an APMSS on how to best optimize message components to improve perceived helpfulness by the users of these systems.

Method

Participants: We recruited 510 participants who scored at least an 8 on the AUDIT screener for a Qualtrics panel study. Participants were recruited through Qualtrics recruitment methods. Participants were required to 1) be 18 years or older, 2) be able to write and read in English, 3) have at least moderately severe alcohol use disorder (>= 4 self-reported DSM-5 symptoms). Participants were determined to be ineligible if they failed an attention check item placed in the survey, or if they completed the survey too quickly. The sample was majority male (61.2%), white (71.0%), and had at least a college education (39.0%).

Procedure: Participants completed a 6-item survey with 7-point Likert-scaled items where they were asked to rate the perceived helpfulness of each tone, and rate preference for informal versus formal messaging. Next, participants completed a separate survey where they were asked to rate the perceived helpfulness of specific messages. In total, participants evaluated 30 messages across the five message tones (legitimizing, self-efficacy, acknowledging, value affirmation and norms), two formality styles (formal; informal), and three simulated user contexts (high, increasing; high, decreasing; low, increasing). User contexts were based on hypothetical outputs from a machine learning model that determines a participant's current daily lapse risk and the change in their lapse risk over the previous two weeks. These messages were generated through the use of Azure OpenAI's GPT 5.1-Chat model, which was given information both from the hypothetical lapse prediction model and from the participants' EMAs. Participants evaluated the helpfulness of each support message on a 7-point Likert scale. There were four survey versions that were distributed evenly across participants, with different variants of each of the messages for each context, tone, and style combination. Simulated contexts and messages within each context were randomized.

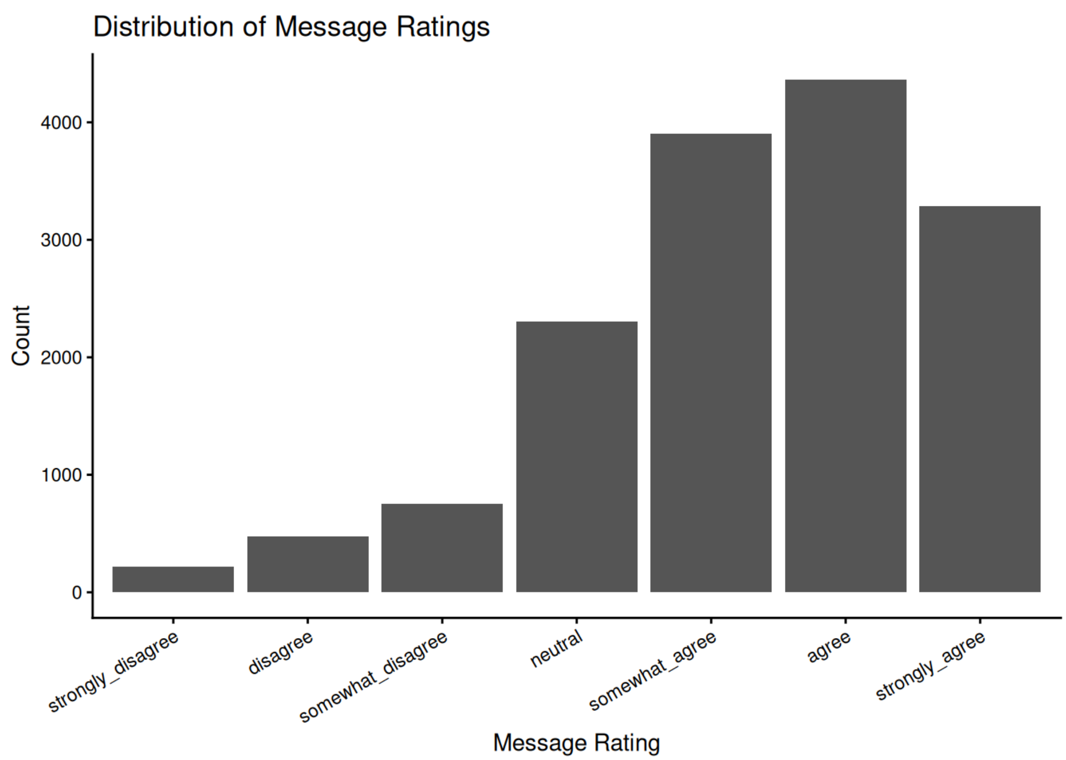
Prompts: The prompts for each of the message tone and style categories are displayed in Appendix A. For each generated message, a simulated user context, message tone, and message style was chosen and given to the LLM. The LLM was also provided with a system prompt that is consistent across all message categories, which is also provided in Appendix A. GPT 5.1-chat was chosen for its relative message quality compared to smaller models such as mistral. GPT 4o was used in testing, but has since been deprecated.

Analyses: Analyses were preregistered on OSF to minimize researcher degrees of freedom (https://osf.io/e34bx/overview). We elected to use an elastic net model because of its simplicity with respect to interpreting feature coefficients. All model configurations varied across appropriate hyperparameters (alpha and lambda) and selection strategies. Posterior probability distributions derived from Bayesian hierarchical generalized linear models were used to compare different selection strategies. Model configurations were trained and evaluated using grouped nested cross-validation, and tone and style selection strategies were evaluated using the same nested cross-validation strategy. Selection strategies (random tone/style, highest tone/style, model-predicted tone/style) were compared using posterior probability distributions derived from a Bayesian hierarchical generalized linear model. Analysis was conducted using R (R Core Team, 2026) and RStudio (Posit Team, 2026).

Results

Figure 1

Overall Message Rating Distribution

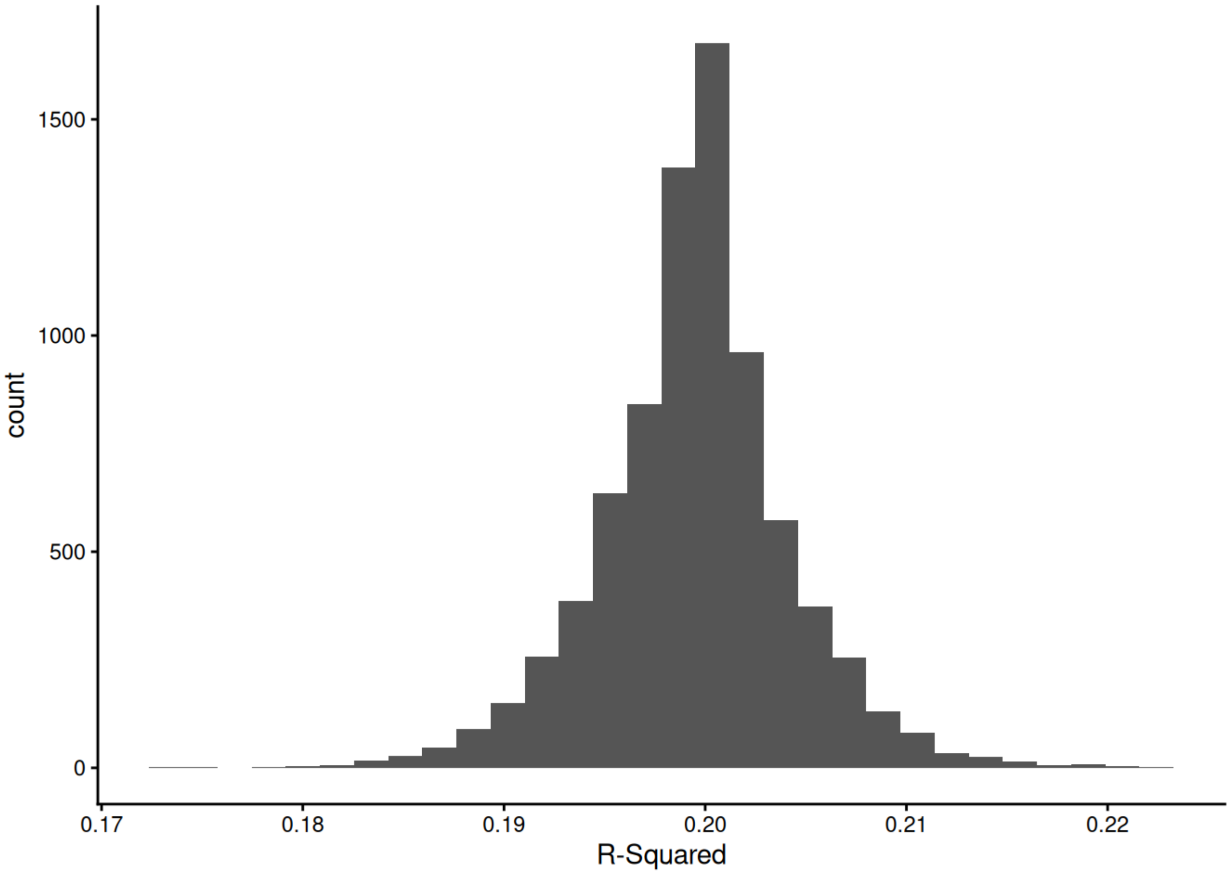


Note. The plot depicts the aggregate distribution of message ratings across all participants. Message ratings ranged from 1-7 on a standard Likert scale anchored at "Strongly Disagree" to "Strongly Agree".

Participants generally found the messages to be helpful, with the overall average message rating occurring between "Somewhat Agree" and "Strongly Agree" (mean = 5.32; Figure 2). There were not significant differences between the tone and style preferences on aggregate, though individual differences were observed, as detailed later.

Figure 2

GLMNet R^2 Distribution

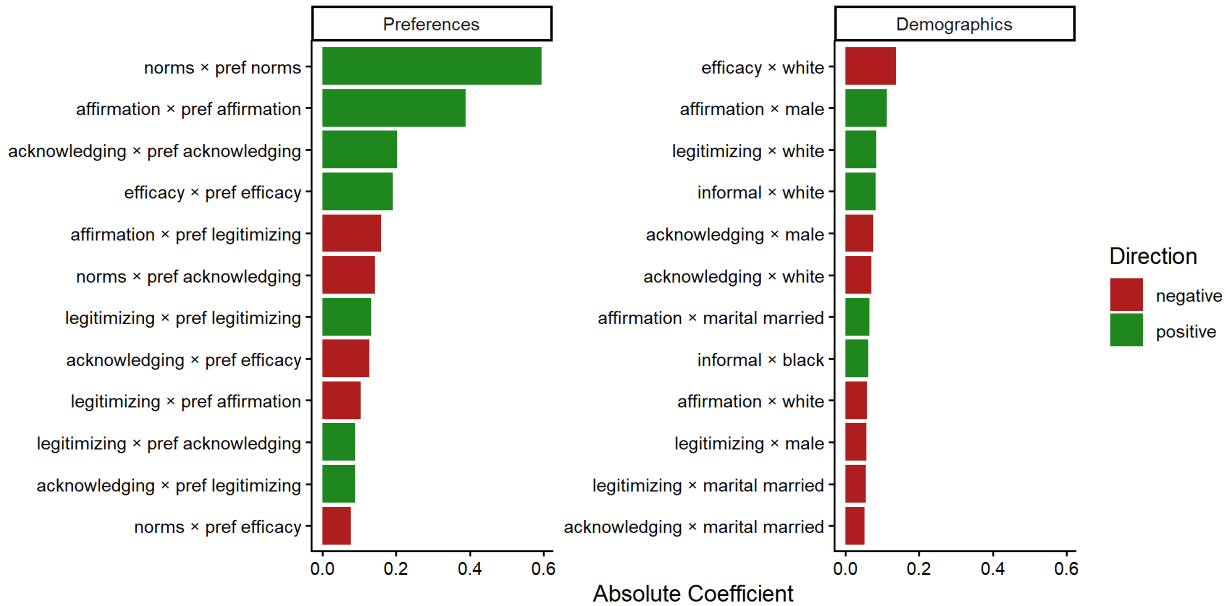


Note. An alpha of 0.1 was chosen for its relative ease of interpretation; future work will likely result in a more performant model.

The selected machine learning model had a R^2 distribution centered around 0.2, which means the model accounts for 20% of the variance (Figure 1).

Figure 3

Message Interaction Effects

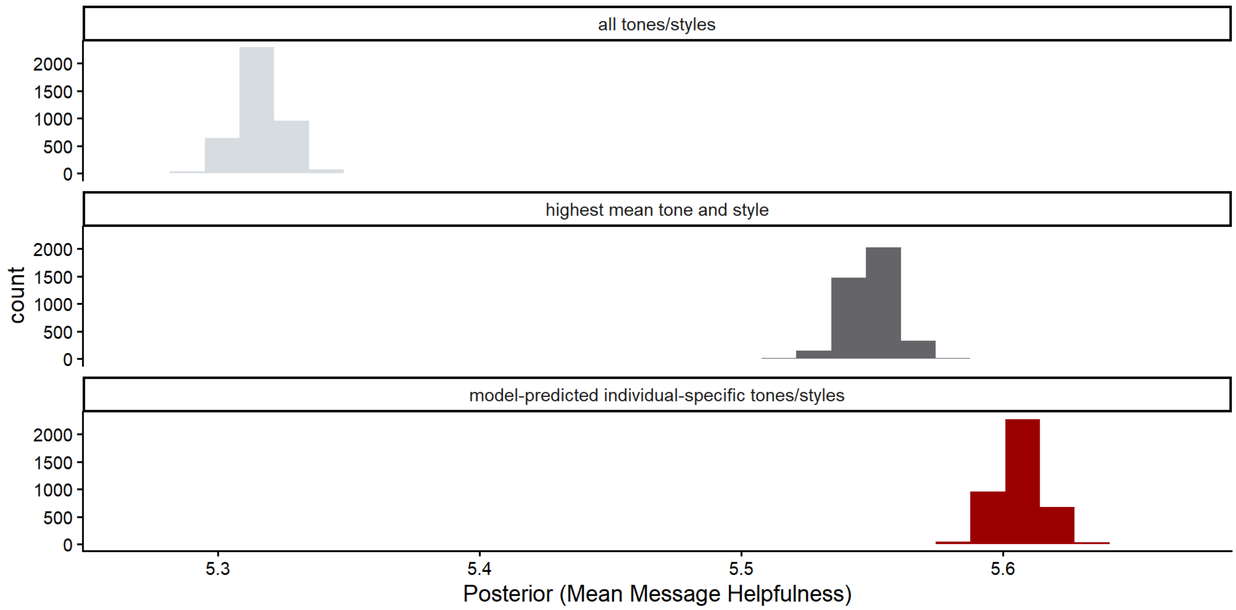


Note. The plot depicts GLMNet coefficients for interactions. Main effects were not included in this plot.

Absolute coefficients were used to examine which features contributed most strongly to model predictions (Figure 3). Interaction effects occurred between participants' stated preferences for each tone and style category and their actual ratings of messages in those categories, e.g. participants with a stated preference for norms messaging tended to rate messages in the norms condition higher. Demographic characteristics exhibited comparably smaller effects.

Figure 4

Posterior Probability Comparison



Note. The plot depicts Bayesian posterior probabilities for each of the three tone and style selection strategies. The top subplot depicts the strategy in which participants receive randomly selected tones and styles, the middle represents strategy in which participants receive the highest overall rated tone and style, and the bottom represents the strategy in which participants receive a model-predicted tone and style.

We used the posterior probability distributions to formally compare the three strategies mentioned above. The median increase for the highest aggregate versus random strategies was 0.234, yielding a probability of 1.000 that the highest aggregate strategy had superior performance. The median increase for the model-predicted strategy versus the highest aggregate strategy was 0.0569, yielding a probability of 1.000 that the model-predicted strategy had superior performance.

Discussion

Implications

Large language models can be used to write daily support messages to scale an automated participant monitoring and support system to meet high needs for continuing care.

Individual model-predicted tones and styles provide the most helpful and possibly equitable messages. Given that the sample consisted of predominantly college-educated white men, writing messages using the highest overall rated tone and style is likely not the most equitable strategy, and using a machine learning model may bridge this gap.

However, use of all tones and styles will provide more variability which may improve engagement for long-term use of a system. In a real world system, such as the one being developed by Wyant et al., long-term engagement is critical for patient outcomes (Wyant et al., 2025).

Future Directions

Further refinement of the model may improve prediction and message tone selection. Additional data cleaning is required to screen out participants that were not removed via passive means (e.g. attention or timing checks). Better model configurations must also be tested; for this preliminary analysis the model was selected for ease of interpretation.

Messages must be evaluated within a live automated monitoring and support system with patients in recovery (Wyant et al., 2025).

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Appendix A: Prompts

Prompt Type	Prompt Text
System; Tone/Style Invariant	“You are a chatbot in an automated recovery support system for people with alcohol use disorder. You are writing a daily message from the recovery support system that will provide the person with information about how they are doing in their recovery today. This information will come from a machine learning model that is run on information provided by the person from a daily survey and their moment-to-moment location. The message should be 3 to 4 sentences long. The message should be written for someone with a 6th grade reading level. Regardless of how formal or informal the user prefers their message to be, the message should appear to have been written by a human. The message should use positive and engaging language that encourages the person to continue to read the daily messages. Do not explicitly greet the person. Do not refer to yourself in the message. Do not provide advice. Avoid the use of words like "should" or "must". Avoid the use of words like "valid" or "heavy". Please avoid using em dashes in your responses. Use commas or parentheses instead. Do not ask questions and do not use exclamation points. Do not imply that ongoing support is available directly from the recovery support system or the developers of the system.”
User; Tone/Style Invariant	"Generate a message for a user based on the following context: <tone prompt> <style prompt> <simulated user context>. This message must inform the user of their lapse risk and lapse risk change (e.g. "your lapse risk is ____ and ____"). Please tailor the message to the user's situation."
Tone; Legitimizing	"The message should let the user know that their feelings and experiences are valid. You should remind them that it is okay to feel this way and not minimize or dismiss their feelings and experiences. You should not make reference to other people's experiences."
Tone; Self-Efficacy	"The message should attempt to make the user feel capable. You should remind them that they have what it takes to keep going and the skills to succeed."
Tone; Acknowledging	"The message should acknowledge the user's feelings and experiences. You should remind them that they have been heard and are encouraged to reflect on their emotions."
Tone; Value Affirmation	"The message should help the user connect with what matters most to them. You should remind them to think about what is important in their life and why they keep going."
Tone; Norms	"The message should help the user remember that others have had similar feelings and experiences. You should remind them that others have had comparable struggles and that theirs are not unusual."

Style; Formal	"Address this user in a formal manner, as if this message were being delivered by a healthcare professional."
Style; Informal	"Address this user in an informal manner, as if this message were being delivered by a peer."